

10 Rafts × 8 Cars = 80 Cars Capacity Industrial Underground Parking System Estimation:

Structure Specifications

Parameter

Value

Number of Rafts

10

Cars per Raft

8

Total Capacity

80 Cars

Raft Diameter

24 m

Raft Height

3 m

Parking Stack Height

30 m

Machine Room

3 m

Foundation

3 m

Total Underground Depth

36 m

Industrial Cost Estimation

Component

Estimated Cost

Excavation & Civil Construction

₹1.5 – 2.5 Crore

RCC Underground Shaft

₹2 – 3 Crore

10 Heavy Duty Circular Rafts

₹1 – 1.5 Crore

Bearings & Rotation Mechanism

₹20 – 40 Lakhs

Hydraulic Lift (350–400 Ton)

₹75 Lakhs – 1.2 Crore

Hydraulic Cylinders

₹30 – 50 Lakhs

Hydraulic Power Pack

₹20 – 30 Lakhs

Motors & Gear Systems

₹20 – 40 Lakhs

PLC Control System

₹15 – 25 Lakhs

IoT Sensors & Monitoring

₹5 – 10 Lakhs

RFID & QR Access System

₹5 – 8 Lakhs

AI Cameras & Security

₹10 – 20 Lakhs

Electrical Infrastructure

₹20 – 40 Lakhs

Fire Safety System

₹15 – 25 Lakhs

Entry & Exit Automation

₹10 – 15 Lakhs

Installation & Testing

₹30 – 50 Lakhs

Total Estimated Cost

Minimum Cost

≈ ₹8 Crore

Maximum Cost

≈ ₹12 Crore